Insertion Sort:

```
def insertionSort(arr):
    n = len(arr)

    if n <= 1:
        return

    # Traverse from second element
    for i in range(1, n):

        # Store current element to be inserted
        key = arr[i]

        # Compare with previous elements
        j = i - 1

        # Shift elements greater than key
        while j >= 0 and key < arr[j]:
            arr[j + 1] = arr[j]
            j -= 1

        # insert key at correct position
        arr[j + 1] = key

arr = [12, 11, 13, 5, -1]
insertionSort(arr)
print(arr)
```

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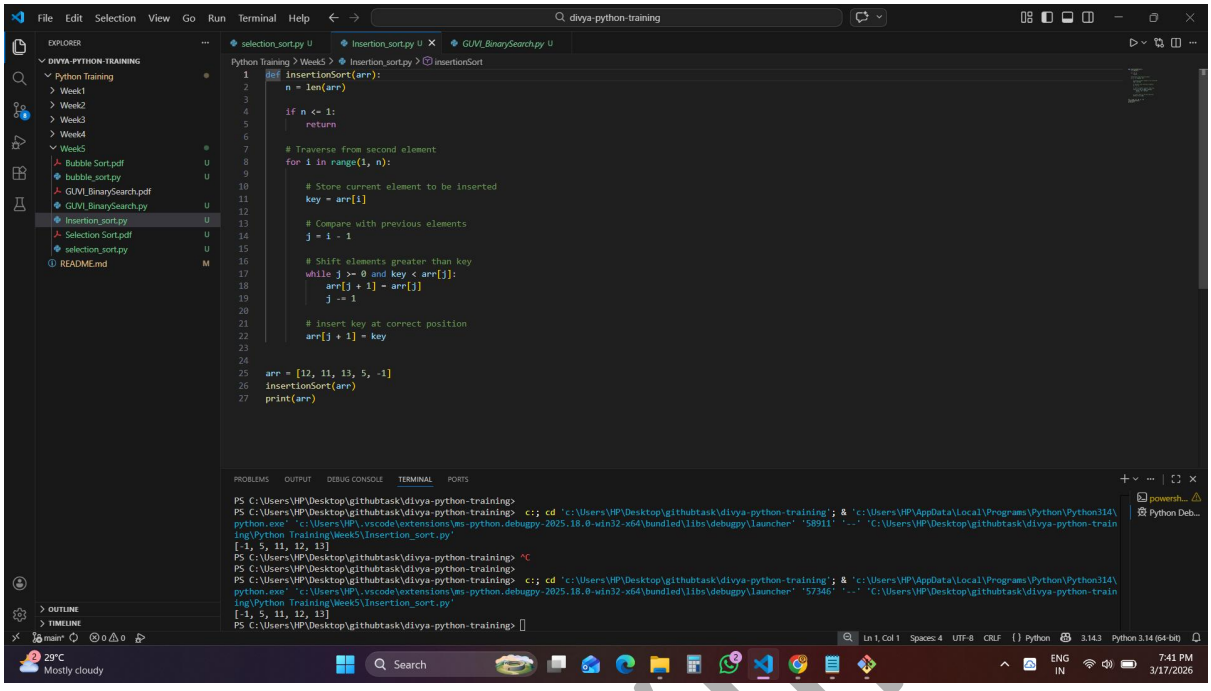
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Week5
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bubble_sort.py
GUVI_BinarySearch.pdf
GUVI_BinarySearch.py
Insertion_sort.py
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selection_sort.py
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Insertion_sort.py U
GUVI_BinarySearch.py U
Python Training > Week5 > Insertion_sort.py > insertionSort
def insertionSort(arr):
n = len(arr)
if n <= 1:
return
# Traverse from second element
for i in range(1, n):
# Store current element to be inserted
key = arr[i]
# Compare with previous elements
j = i - 1
# Shift elements greater than key
while j >= 0 and key < arr[j]:
arr[j + 1] = arr[j]
j -= 1
# insert key at correct position
arr[j + 1] = key
arr = [12, 11, 13, 5, -1]
insertionSort(arr)
print(arr)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\HP\Desktop\githubtask\divya-python-training>
PS C:\Users\HP\Desktop\githubtask\divya-python-training> c:; cd 'c:\Users\HP\Desktop\githubtask\divya-python-training'; & 'c:\Users\HP\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58911' '--' 'C:\Users\HP\Desktop\githubtask\divya-python-training\Python Training\Week5\Insertion_sort.py'
[-1, 5, 11, 12, 13]
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[-1, 5, 11, 12, 13]
PS C:\Users\HP\Desktop\githubtask\divya-python-training>
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